That implies that, v_{TRANS} and v_{ROT} are equal in magnitude and opposite in direction ($v = v_{TRANS} - v_{ROT} = 0$). Hence, we conclude that in pure rolling, for all the points on the edge, the magnitudes of v_{TRANS} and v_{ROT} are equal ($v_{TRANS} = v_{ROT}$). As $v_{TRANS} = v_{CM}$ and $v_{ROT} = R\omega$, in pure rolling we have,

$$v_{CM} = R\,\omega \qquad (5.55)$$

We should remember the special feature of the equation 5.55. In rotational motion, as per the relation $v = r\omega$, the centre point will not have any velocity as r is zero. But in rolling motion, it suggests that the centre point has a velocity v_{CM} given by equation 5.55.

For the topmost point, the two velocities v_{TRANS} and v_{ROT} are equal in magnitude and in the same direction (to the right). Thus, the resultant velocity v is the sum of these two velocities, $v = v_{TRANS} + v_{ROT}$. In other form, $v = 2\,v_{CM}$ as shown in Figure 5.34.

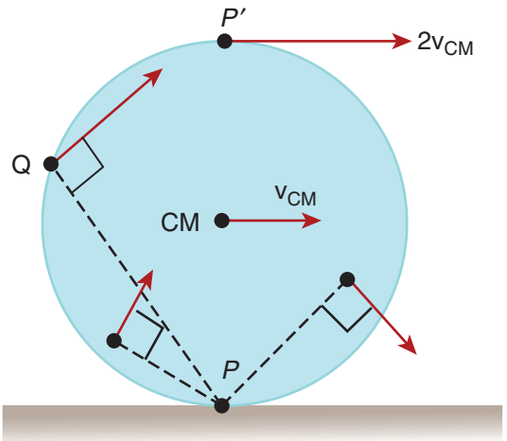
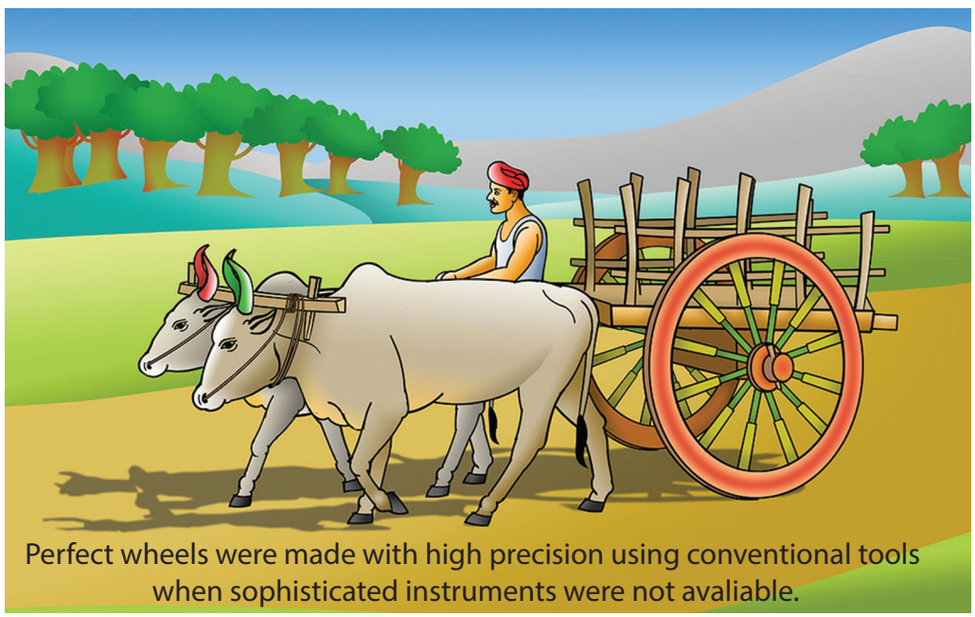


Figure 5.34 Velocity of different point in pure rolling

5.6.2 Slipping and Sliding

When the round object moves, it always tends to roll on any surface which has a coefficient of friction any value greater than zero ($\mu > 0$). The friction that enabling the rolling motion is called rolling friction. In pure rolling, there is no relative motion of the point of contact with the surface. When the rolling object speeds up or slows down,



Perfect wheels were made with high precision using conventional tools when sophisticated instruments were not avaliable.

UNIT-5(XI-Physics_Vol-1).indd 253 06-01-2022 17:48:03